llmXive Automated Review of SciAtlas: A Large-Scale Knowledge Graph for Automated Scientific Research

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is **advisory, automated feedback for the authors**: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The manuscript presents a large-scale knowledge graph (SciAtlas) and a retrieval pipeline. While the core concept is sound, there are significant factual inconsistencies regarding the dataset statistics and citation accuracy that must be resolved before acceptance.

1. Contradictory Statistics: There is a direct conflict between the detailed statistics provided in the text and the summary table in the second chunk (e001). - In **e000** (Section "Statistics" and Table 1), the paper states there are **213.88M** 'CITES' relations and **2.06B** 'COAUTHOR' relations. - In **e001** (Table 1), the same table lists **1.2B** 'CITES' and **800M** 'COAUTHOR'. These numbers are not just slightly different; they are orders of magnitude apart and contradict each other directly. The 'COAUTHOR' count in e001 (800M) is less than half the count in e000 (2.06B), while the 'CITES' count in e001 (1.2B) is nearly six times higher than in e000 (213.88M). The authors must unify these figures across the manuscript.

2. "3 Billion Triplets" Claim: The abstract and introduction claim the graph contains "3B triplets" (edges). However, summing the specific edge counts provided in the detailed table in **e000** (213.88M + 101.38M + 105.89M + 149.00M + 195.94M + 2.06B + 60.37M + 252 + 68.38M + 40.90M + 26) yields approximately **2.83B**. While "3B" is a reasonable rounded figure, the discrepancy with the contradictory numbers in e001 (where the sum would be vastly different) makes the claim ambiguous. The authors should clarify if the 3B figure is an upper bound, a rounded total, or if the detailed counts are missing a significant category.

3. Citation Validity (Qwen3): The paper cites 'Qwen3-30B-A3B-Instruct' (e001) and 'Qwen3' (e000) as the keyword extraction model. As of the current date, the Qwen3 series has not been publicly released (Qwen2.5 is the latest stable version). Unless this is a specific internal or pre-release model with a verifiable technical report, this citation is likely hallucinated or premature. The authors must verify the model name and provide a valid citation or correct the model version to the actual one used (e.g., Qwen2.5).

4. Data Source Availability: The bibliography entry for OpenAlex (c-001) is flagged as "unreachable". While external links can rot, the authors should ensure the primary data source link is stable or provide a DOI/Archive link for the specific data snapshot used to construct the graph to ensure reproducibility.

These issues are primarily writing and factual consistency errors that can be fixed by cross-referencing the data generation logs and updating the text/tables accordingly.

- [**writing**] Table 1 (e001) lists CITES edges as 1.2B and COAUTHOR as 800M, but Table 1 (e000) and the text (e000) state 213.88M CITES and 2.06B COAUTHOR. These are direct contradictions. The authors must reconcile these numbers and ensure the table matches the text.
- [**writing**] The abstract and introduction claim the KG contains '3B triplets' (edges), but the sum of the specific edge counts provided in the detailed table (e000) is approximately 2.8B. The authors should clarify if '3B' is a rounded estimate or if the detailed counts are incomplete.
- [**writing**] The paper cites 'Qwen3-30B-A3B-Instruct' (e001) and 'Qwen3' (e000). As of the current date, Qwen3 is not a publicly released model family (Qwen2.5 is the latest). The authors must verify the model name/version or provide a citation if this is a pre-release/internal model.
- [**writing**] The bibliography lists the OpenAlex URL (c-001) as 'unreachable'. While external links can change, the authors should verify the stability of the data source link or provide a DOI/Archive link for the specific snapshot used.

paper_reviewer_figure_critic

Verdict: minor_revision
Figure 1
The figure effectively visualizes the distribution of disciplines, but the caption contains a grammatical error with a missing subject, and the chart's smaller slices are too crowded to read clearly.

Figure 2
The figure effectively visualizes the knowledge graph schema with clear entity boxes and relationship arrows. However, the caption is grammatically broken due to missing text, and several attribute labels within the diagram are illegible due to truncation.

- [**writing**] Figure 1: The caption contains a grammatical error and missing subject ('Discipline Distribution in . is a large-scale...'), likely due to a placeholder variable not being replaced with the system name.

- [**writing**] Figure 1: The legend lists 26 disciplines, but the pie chart labels are crowded and illegible for the smaller slices (e.g., <2%), making it difficult to map specific colors to the legend entries without zooming.
- [**writing**] Figure 2: The caption contains multiple instances of missing text where the system name should appear (e.g., 'Schema of .', 'provides a structured...', 'of can be found in Appx..'). This makes the figure description grammatically incomplete and unclear.
- [**writing**] Figure 2: Several attribute labels in the 'Author' and 'Institution' entity boxes are truncated with ellipses (e.g., 'displ...', 'works_co...', 'cited_by_c...'), reducing readability and preventing verification of the full schema.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript relies heavily on high-level, abstract terminology that obscures the specific technical contributions for a general scientific audience. The most significant issue is the overuse of "neuro-symbolic" (Abstract, Sec 1, Sec 3) without a clear, plain-English definition of how the neural (vector embeddings) and symbolic (graph logic) components interact. While experts in the field may recognize the term, a broader audience requires an explicit explanation of the mechanism.

Similarly, phrases like "topological reasoning" (Abstract, Intro) and "deterministic association discovery" (Abstract, Conclusion) are used as buzzwords. "Topological" is a specific mathematical concept; here, it likely just means "structural" or "relational." Using the more precise "graph-based reasoning" would be clearer. The term "tri-path collaborative recall" (Sec 3) is introduced without defining the three paths until later in the section, forcing the reader to guess the meaning.

Metaphorical language further clouds the text. "Cognitive substrate" (Abstract, Sec 2) is a vague metaphor for a database or knowledge base. "Panoramic scientific evolution network" (Abstract) is flowery; "comprehensive map of scientific progress" conveys the same meaning more directly. In the "Idea Grounding" example, the verb "fossilize" is used to describe bias entrenchment; "entrench" or "reinforce" is standard and clearer. Finally, "atomic knowledge" (Limitations) is undefined; specifying "individual facts" or "discrete data points" would prevent ambiguity.

The paper also uses "cognitive map" (Abstract, Conclusion) as a metaphor for the retrieval system. While evocative, it should be accompanied by a literal description (e.g., "a structured representation of relationships") to ensure non-specialists understand the system's actual function. Reducing this density of jargon and metaphors will make the paper's contributions accessible to a wider scientific community.

- [**writing**] The manuscript relies heavily on high-level, abstract terminology that obscures the specific technical contributions for a general scientific audience. The most significant issue is the overuse of "neuro-symbolic" (Abstract, Sec 1, Sec 3) without a clear, plain-English definition of how the neural (vector embeddings) and symbolic (graph logic) components interact. While experts in the field may recognize the term, a broader audience requires an explicit explanation of the mechanism. Similarly, p

paper_reviewer_logical_consistency

Verdict: minor_revision

The manuscript presents a large-scale knowledge graph and a retrieval pipeline, but several internal inconsistencies and unsupported causal claims weaken the logical consistency of the conclusions.

First, there is a severe data contradiction regarding the graph statistics. In Section 2 (Overview) and Table 1 (e000), the authors state there are 213.88M CITES edges and 2.06B COAUTHOR edges. However, the summary Table 1 in Section 2 (e001) lists CITES as 1.2B and COAUTHOR as 800M. This direct numerical conflict creates a logical impossibility: the graph cannot simultaneously have these two different sets of statistics. This undermines the credibility of the "large-scale" claim and the subsequent reasoning based on these numbers.

Second, the terminology "tri-path collaborative recall" (Abstract, Section 3) is not logically aligned with the described mechanism. The authors define three matching paths (Keyword, Semantic, Title) in Section 3.1, but these are used to generate a unified seed set ($\mathcal{P}_{seed}$) via a weighted sum (Eq. 5). The subsequent "graph propagation" (Section 3.3) is a single Random Walk with Restart on this unified seed set, not a parallel traversal of three distinct paths. The conclusion that the system achieves "tri-path" reasoning is a non-sequitur; the mechanism is actually "multi-source seed fusion followed by single-path diffusion."

Finally, the claim that the system "reduces inference costs" (Abstract, Conclusion) lacks a supporting premise. The proposed pipeline involves multiple LLM calls (keyword extraction, reranking) and graph computations. Without a comparative experiment showing lower token usage or latency than a baseline (e.g., standard vector search), the causal link between the proposed architecture and cost reduction is unsupported. The argument assumes that "structured reasoning" is inherently cheaper, which is not necessarily true given the overhead of graph traversal and LLM-based preprocessing.

- [**science**] Resolve the contradiction in Table 1 (e001) where CITES edges are listed as 1.2B and COAUTHOR as 800M, conflicting with the text and Table 1 (e000) stating 213.88M CITES and 2.06B COAUTHOR. This data swap undermines the statistical validity of the graph description.
- [**writing**] Clarify the 'tri-path' claim. The text describes three matching paths (Sec 3.1) but merges them into a single seed set before a single Random Walk (Sec 3.3). The term implies parallel graph exploration, but the mechanism is sequential filtering. Re-evaluate terminology to match the algorithm.
- [**science**] The claim that the system reduces inference costs (Abstract, Conclusion) lacks support. The pipeline involves multiple LLM calls and graph traversal. Without a comparative analysis of token usage or latency against a baseline, the causal claim of cost reduction is unsupported.

paper_reviewer_overreach

Verdict: minor_revision

The manuscript makes several claims that extend beyond the evidence provided in the text, particularly regarding the nature of the retrieval algorithm and the comparative benefits of the system.

First, the Abstract and Conclusion repeatedly describe the retrieval pipeline as achieving "deterministic association discovery." This is a significant overreach. The core algorithm employed is Random Walk with Restart (RWR), which is inherently a stochastic process. While the implementation yields reproducible results for fixed inputs, characterizing the discovery mechanism as "deterministic" misrepresents the probabilistic nature of graph diffusion and suggests a level of logical certainty that the method does not possess. This phrasing should be corrected to "probabilistic topological reasoning" or "stochastic association discovery" to maintain scientific accuracy.

Second, the paper asserts that SciAtlas "reduces reasoning costs" (Abstract, Conclusion) compared to existing agentic frameworks. However, the manuscript provides no empirical data to support this. While a runtime of " 2 minutes" is mentioned, there is no baseline comparison against standard vector retrieval or multi-hop agentic search in terms of token consumption, API latency, or computational overhead. Without a controlled experiment or at least a theoretical breakdown of cost savings, this claim remains an unsupported extrapolation.

Third, the Limitations section makes the absolute claim that "KG is indispensable for scientific discovery." This is a philosophical stance rather than a conclusion drawn from the paper's data. The authors demonstrate that their KG is *useful*, but they have not conducted a study proving that non-KG approaches are fundamentally incapable of scientific discovery. This absolute language should be tempered to reflect the paper's actual contribution, such as "KGs provide a structured substrate that enhances..."

Finally, the claim that the system "dismantles disciplinary barriers" (Abstract, Introduction) is contradicted by the provided statistics. The graph is heavily dominated by Medicine (18.56%) and Social Sciences (10.70%), while Computer Science represents only 6.29% of the corpus. This distribution suggests the system may currently reflect existing publication biases rather than effectively dismantling them. The claim should be qualified to acknowledge the current data distribution or removed until a more balanced corpus is demonstrated.

- [**writing**] The manuscript makes several claims that extend beyond the evidence provided in the text, particularly regarding the nature of the retrieval algorithm and the comparative benefits of the system. First, the Abstract and Conclusion repeatedly describe the retrieval pipeline as achieving "deterministic association discovery." This is a significant overreach. The core algorithm employed is Random Walk with Restart (RWR), which is inherently a stochastic process. While the implementation yields repro

paper_reviewer_safety_ethics

Verdict: minor_revision
The manuscript presents SciAtlas, a large-scale knowledge graph integrating over 43 million papers and 157 million entities, including detailed author and institutional data. While the technical contribution is significant, the paper currently lacks sufficient discussion on the ethical implications and safety risks associated with the scale and application of this system.

First, the **Data Privacy and Consent** aspect is under-addressed. The dataset includes 109.7 million author nodes and 195.94 million affiliation edges (Table 1, Sec 2.1). Although the data is sourced from OpenAlex, the manuscript does not explicitly state how the authors handle privacy concerns regarding individual researchers, particularly in the context of the "Researcher Background Review" application (Sec 5.6). There is no mention of compliance with GDPR, CCPA, or the specific terms of service of the source data regarding the aggregation and summarization of personal academic profiles. An explicit Ethics Statement or Data Privacy section is required to confirm that the data usage aligns with the source's licensing and that no sensitive personal information is being exposed or misused.

Second, the **Dual-Use and Safety Risks** of the "Idea Generation" module (Sec 5.3) are not evaluated. The system is designed to "relax constraints on distant nodes" to explore interdisciplinary concepts. While this fosters innovation, it also lowers the barrier for generating novel research ideas that could be misused for harmful purposes (e.g., synthesizing pathogens, designing cyber-attacks, or creating disinformation campaigns). The paper currently frames this capability purely as a benefit without acknowledging the potential for misuse or proposing any safety guardrails (e.g., filtering mechanisms for high-risk topics). A discussion on these risks and the authors' mitigation strategies is necessary.

Finally, the **Bias and Fairness** implications of the "Researcher Background Review" (Sec 5.6) and the underlying graph construction need attention. The system relies on citation counts and LLM-generated summaries to profile researchers. This could inadvertently amplify existing biases in the scientific literature (e.g., gender, geographic, or institutional biases) if the LLM summarization or the graph weighting (e.g., Eq. 6-7) favors established, high-citation entities over emerging or underrepresented voices. The manuscript should address how these potential biases are monitored or mitigated to prevent the system from reinforcing inequities in the scientific community.

In summary, while the system is technically sound, the lack of an explicit ethics statement, a discussion on dual-use risks, and an analysis of bias in researcher profiling constitutes a gap in the current manuscript.

- [**writing**] The manuscript lacks a dedicated Ethics Statement or Data Privacy section. Given the inclusion of 109.7M author nodes and 195.94M affiliation edges, explicit confirmation of compliance with OpenAlex's Terms of Service and GDPR/CCPA regarding researcher data is required.
- [**writing**] The 'Idea Generation' application (Sec 5.3) proposes using the KG to synthesize novel research concepts. A discussion on the risks of dual-use (e.g., accelerating the discovery of harmful biological agents or cyber-attack vectors) and the system's potential to automate

the generation of disinformation or biased scientific claims is missing.

- [**writing**] The 'Researcher Background Review' feature (Sec 5.6) aggregates and summarizes individual academic trajectories. The paper must clarify the consent model for this profiling and address potential biases in the LLM-generated summaries that could unfairly impact researchers' reputations.

paper_reviewer_scientific_evidence

Verdict: minor_revision
The manuscript presents SciAtlas, a large-scale knowledge graph, and a neuro-symbolic retrieval pipeline. While the scale of the data (43M papers, 3B edges) is impressive, the scientific evidence supporting the *efficacy* of the proposed retrieval method is currently insufficient.

The central claim is that the "neuro-symbolic retrieval algorithm" (combining keyword, semantic, and graph-based RWR) outperforms or meaningfully improves upon existing vector/keyword matching. However, Section 4 (Neuro-Symbolic Retrieval) and Section 5 (Downstream Applications) describe the *mechanism* and provide qualitative examples (e.g., "An Example of Idea Grounding") but lack any quantitative evaluation. There are no reported metrics such as Precision@K, Recall, Mean Average Precision (MAP), or NDCG on a standard benchmark or a held-out test set. Without these metrics, it is impossible to verify if the complex weighting scheme (Eqs. 1-10) actually yields better results than a simple vector search baseline. The claim of "deterministic association discovery" is a qualitative assertion, not a scientifically validated result.

Furthermore, the construction of the graph relies heavily on an LLM (Qwen3) for keyword extraction (Section 3.2). The quality of these 3.76M keyword nodes is fundamental to the graph's topology. The paper provides no evidence of the extraction quality, such as human evaluation scores, inter-annotator agreement, or comparison against a gold-standard keyword set. If the keyword extraction is noisy, the "semantic level" and "conceptual level" of the graph may be compromised, invalidating the downstream retrieval performance.

Finally, the retrieval algorithm introduces numerous hyperparameters (e.g., $\theta_{kw}=0.7$, $\lambda_{emb}=0.3$, $\gamma=0.5$, α in RWR). The manuscript does not describe how these were tuned. There is no mention of an ablation study to determine the contribution of each component (e.g., does the RWR actually add value over the seed matching?). The lack of sensitivity analysis or parameter justification suggests the results may be fragile or overfit to the specific examples shown. To support the scientific claims, the authors must provide quantitative benchmarks, validation of the data construction pipeline, and an analysis of the algorithm's hyperparameters.

- [**science**] The paper claims a 'neuro-symbolic retrieval algorithm' achieves 'deterministic association discovery' but provides no quantitative evaluation (e.g., precision@k, recall, NDCG) against baselines. Without empirical metrics on a held-out test set, the efficacy of the proposed RWR and weighting scheme remains an unverified assertion.
- [**science**] The construction pipeline relies on a single LLM (Qwen3) for keyword extraction

without reporting inter-annotator agreement, human validation, or error rates. The quality of the 3.76M keyword nodes is a critical dependency for the graph's utility, yet no evidence of extraction accuracy is provided.

- [**science**] The retrieval algorithm contains multiple tunable hyperparameters (e.g., theta_kw=0.7, lambda_emb=0.3, gamma=0.5). The manuscript does not describe the methodology used to select these values (e.g., grid search, ablation study) or report sensitivity analysis, raising concerns about overfitting to anecdotal examples.

paper_reviewer_statistical_analysis

Verdict: minor_revision
The manuscript presents a large-scale knowledge graph (SciAtlas) and a neuro-symbolic retrieval pipeline. However, from a statistical analysis perspective, the paper lacks the necessary quantitative rigor to support its claims of efficacy and novelty.

First, the retrieval algorithm relies heavily on a set of fixed hyperparameters (e.g., $\gamma=0.5$ in Eq. 7, $\beta_{cite}=1.00$ in Table 1, λ weights in Eq. 10). The manuscript provides no statistical justification for these specific values. There is no sensitivity analysis, ablation study, or cross-validation reported to demonstrate that these parameters are optimal or robust. Without this, the system's performance appears arbitrary rather than empirically derived.

Second, and most critically, the paper makes strong claims about the effectiveness of the retrieval method ("seamless transition," "reducing inference cost") but presents **no statistical evaluation results**. There are no tables or figures reporting standard information retrieval metrics such as Precision@K, Recall@K, NDCG, or Mean Average Precision (MAP) on a held-out test set. Furthermore, there is no comparison against baseline methods (e.g., pure vector search or keyword search) with statistical significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) to validate that the proposed method outperforms existing approaches. The absence of confidence intervals or error bars on any reported metrics (if they were present in the unseen figures) would also be a concern, but the total lack of quantitative performance data is the primary deficit.

Finally, the construction of the graph relies on an LLM (Qwen3) to extract keywords and assign importance scores. The reliability of these features is crucial for the downstream graph traversal. The manuscript does not report any validation statistics, such as inter-annotator agreement (Cohen's Kappa) or correlation with human-annotated ground truth, to assess the quality and consistency of the LLM-generated features. Without this validation, the statistical soundness of the graph's semantic layer is questionable.

To be accepted, the authors must provide a rigorous statistical evaluation of their retrieval system, including baseline comparisons, significance testing, and sensitivity analysis of the hyperparameters.

- [**science**] The manuscript defines edge weights (Table 1) and node importance scores (Eq. 6-7) using fixed hyperparameters (e.g., gamma=0.5, beta_cite=1.00) without reporting any sensitivity analysis, ablation studies, or statistical justification for these specific values. The impact of these choices on retrieval performance is unquantified.

- [**science**] The paper claims a 'neuro-symbolic retrieval algorithm' but provides no statistical evaluation metrics (e.g., Precision@K, Recall@K, NDCG, or Mean Average Precision) comparing the proposed method against baselines. Without quantitative results and confidence intervals, the claim of 'seamless transition' and 'reduced inference cost' is unsupported by evidence.
- [**science**] The construction pipeline relies on LLMs (Qwen3) for keyword extraction and scoring. The manuscript fails to report inter-annotator agreement (e.g., Cohen's Kappa) or validation statistics against a human-annotated gold standard to assess the reliability and bias of these extracted features.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript presents a compelling system but suffers from significant inconsistencies in naming and data presentation that undermine its professional polish.

First, there is a critical inconsistency in the system's name. While the title and most of the text refer to "SciAtlas," the caption for Table 1 in section e001 explicitly states: "SciMap comprises a total node count..." This suggests a copy-paste error from a previous draft or a different project. The name must be standardized to "SciAtlas" throughout the entire document, including all figure captions, table headers, and the abstract.

Second, the data presented in the tables is inconsistent. In section e000, Table 1 lists the "CITES" relation count as 213.88M and "COAUTHOR" as 2.06B. However, in section e001, the corresponding table lists "CITES" as 1.2B and "COAUTHOR" as 800M. These are mutually exclusive values. The authors must verify the correct statistics and ensure they are identical in every instance where they appear.

Third, the writing quality in the abstract and introduction requires refinement. The abstract contains a sentence fragment: "Agentic deep-research-based frameworks are often prone to logical hallucinations and consuming high inference costs." The phrase "consuming high inference costs" is grammatically disconnected from the subject. It should be rephrased, for example: "...and often incur high inference costs." Additionally, the phrase "In this report, we introduce" is slightly informal for a research paper; "In this paper" or "We present" is preferred.

Finally, there are minor LaTeX and formatting issues. In the table in e001, the use of "\dots" in the "Statistics" section is acceptable for brevity, but the caption's reference to "SciMap" must be fixed. The authors should also ensure that all variable definitions (e.g., λ_{emb}, λ_{title}) are introduced consistently before they are used in equations.

Addressing these inconsistencies and grammatical errors is essential before the paper can be considered for publication.

- [**writing**] Inconsistent naming: The paper alternates between 'SciAtlas' and 'SciMap' (e.g., Table 1 caption in e001). Standardize to 'SciAtlas' throughout.

- **[writing]** Grammar and flow: The abstract contains a sentence fragment ('...consuming high inference costs') and awkward phrasing ('In this report, we introduce...'). Revise for professional academic tone.
- **[writing]** Table inconsistency: Table 1 in e001 lists 'CITES' as 1.2B and 'COAUTHOR' as 800M, while the text and Table 1 in e000 list different values (213.88M CITES, 2.06B COAUTHOR). Ensure data consistency across all tables.
- **[writing]** LaTeX syntax error: In e001, the table caption for 'Statistics of SciAtlas' contains a typo 'SciMap' instead of 'SciAtlas'. Additionally, the table body uses '...' placeholders which should be replaced with actual data or removed.